

```
..     for i in range(0, len(text), 2):
..         a, b = text[i], text[i + 1]
..         r1, c1 = find_position(matrix, a)
..         r2, c2 = find_position(matrix, b)
..
..         if r1 == r2:
..             cipher += matrix[r1][(c1 + 1) % 5]
..             cipher += matrix[r2][(c2 + 1) % 5]
..         elif c1 == c2:
..             cipher += matrix[(r1 + 1) % 5][c1]
..             cipher += matrix[(r2 + 1) % 5][c2]
..         else:
..             cipher += matrix[r1][c2]
..             cipher += matrix[r2][c1]
..
..     return cipher
..
.. key = input("Enter keyword: ")
.. plaintext = input("Enter plaintext: ")
..
.. matrix = generate_matrix(key)
..
.. print("\nPlayfair Matrix:")
.. for row in matrix:
..     print(" ".join(row))
..
.. cipher = encrypt(matrix, plaintext)
..
.. print("\nEncrypted Text:", cipher)
..
Enter keyword: MONARCHY
Enter plaintext: HELLO
```

Playfair Matrix:

M	O	N	A	R
H	Y	B	D	
E	F	G	I	K
L	P	Q	S	T
J	V	W	X	Z

Encrypted Text: CFSUPM